

Land Survey

Engineering Mathematics I

Learner Notes

Foundational mathematics for field measurements, coordinates, bearings, and survey calculations.

Learning outcome	Focus
Apply arithmetic and algebra	Applied mathematics in workplace and training contexts.
Use trigonometric ratios	Applied mathematics in workplace and training contexts.
Calculate distances and bearings	Applied mathematics in workplace and training contexts.

1. Topic Overview

This topic develops mathematical skills required in Land Survey. Learners should show clear method, correct units, and technical interpretation.

2. Key Formulas

$$d = \sqrt{(\Delta E)^2 + (\Delta N)^2}$$

$$\theta = \tan^{-1} \left(\frac{\Delta E}{\Delta N} \right)$$

$$\sin \theta = \frac{O}{H}$$

$$\cos \theta = \frac{A}{H}$$

3. Worked-Example Method

Step 1: identify the given data. Step 2: select the correct formula. Step 3: substitute values carefully. Step 4: compute. Step 5: interpret the answer with correct units.

$$d = \sqrt{(\Delta E)^2 + (\Delta N)^2}$$

4. Practice Questions

1. Find distance between two survey points.

2. Calculate bearing from coordinate differences.
3. Use trigonometry to calculate height.